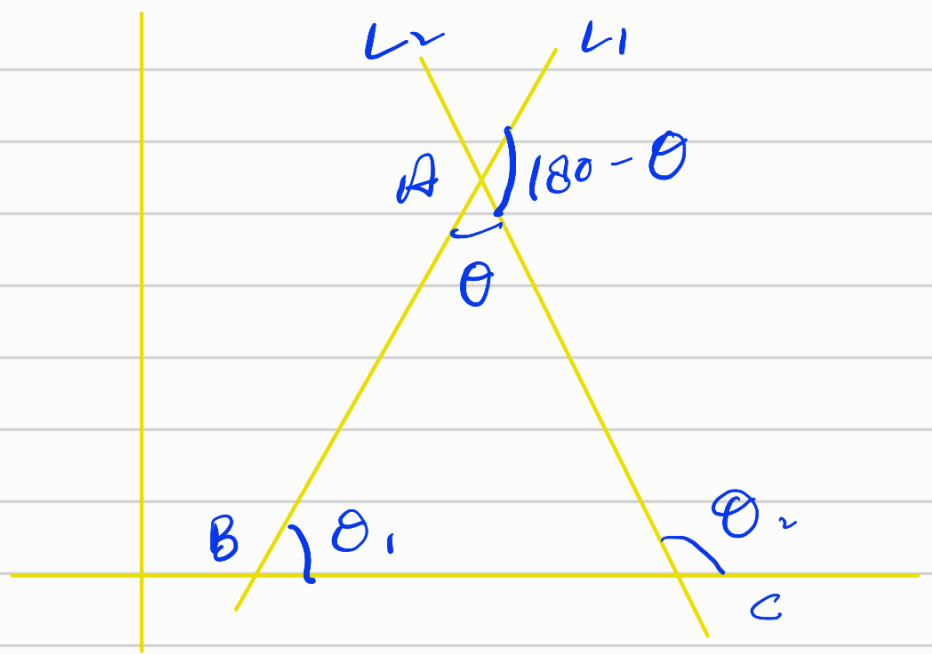




Slope of $L_1 = \tan \theta_1 = M_1$

Slope of $L_2 = \tan \theta_2 = M_2$

In $\triangle ABC$

$\theta_2 = \theta + \theta_1$

(as exterior angle is equal to sum of 2 interior angles)

$\theta = \theta_2 - \theta_1$

Taking $\tan$ on b/s

$\Rightarrow \tan \theta = \tan (\theta_2 - \theta_1)$

$\left\{ \tan (a-b) = \frac{\tan a - \tan b}{1 + \tan a \tan b} \right\}$

$\Rightarrow \tan \theta = \frac{\tan \theta_2 - \tan \theta_1}{1 + \tan \theta_2 \cdot \tan \theta_1}$

putting values of $\tan \theta_1$ and $\tan \theta_2$

$$\Rightarrow \tan \theta = \frac{M_2 - M_1}{1 + M_1 M_2}$$

$\tan \theta$ can some times negative

$$\therefore \tan \theta = \left| \frac{M_2 - M_1}{1 + M_1 M_2} \right|$$

When lines are parallel

Slope of $L_1 = \tan \theta_1 = M_1$ ————— L_1

Slope of $L_2 = \tan \theta_2 = M_2$ ————— L_2

we know that

$$\theta = 0^\circ$$

$$\tan \theta = \left| \frac{M_1 - M_2}{1 + M_1 M_2} \right|$$

$$\tan 0^\circ = \left| \frac{M_1 - M_2}{1 + M_1 M_2} \right|$$

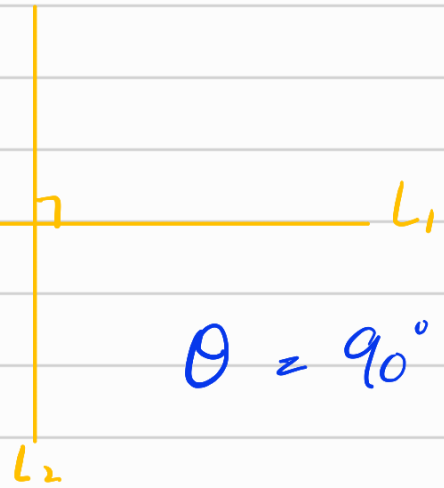
$$\Rightarrow 0 = \left| \frac{M_1 - M_2}{1 + M_1 M_2} \right| = M_1 - M_2 = 0$$

$$\Rightarrow M_1 = M_2$$

when lines are $\perp$ to each other

$$\text{Slope of } L_1 = \tan \theta_1 = M_1$$

$$\text{Slope of } L_2 = \tan \theta_2 = M_2$$



$$\tan 90^\circ = \left| \frac{M_1 - M_2}{1 + M_1 M_2} \right|$$

$$\rightarrow \frac{1}{\tan 90^\circ} = \left| \frac{1 + M_1 M_2}{M_1 - M_2} \right|$$

$$\rightarrow \frac{1}{\infty} = \left| \frac{1 + M_1 M_2}{M_1 - M_2} \right|$$

$$\rightarrow 0 = \frac{1 + M_1 M_2}{M_1 - M_2}$$

$$\rightarrow 1 + M_1 M_2 = 0$$

$$\rightarrow \boxed{M_1 M_2 = -1}$$